

# EXP 1

## Aim

To read an image using Python and convert the image into grayscale.

## Algorithm

1. Import the OpenCV (cv2) library.
2. Read the input image using cv2.imread().
3. Convert the image from color to grayscale using cv2.cvtColor().
4. Display the original image.
5. Display the grayscale image.
6. Wait for a key press and close all display windows.

## Code

```
1  import sys
2  from pathlib import Path
3  import cv2
4
5
6  def process_image(input_path: str, show_windows: bool = False) -> None:
7      image = cv2.imread(input_path)
8
9      if image is None:
10         raise FileNotFoundError(f"Could not read image: {input_path}")
11
12     grayscale_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
13
14     input_file = Path(input_path)
15     grayscale_path = input_file.with_name(f"{input_file.stem}_grayscale{input_file.suffix}")
16
17     cv2.imwrite(str(grayscale_path), grayscale_image)
18
19     print(f"Input image read from: {input_path}")
20     print(f"Grayscale image saved as: {grayscale_path}")
21
22     if show_windows:
23         cv2.imshow("Original Image", image)
24         cv2.imshow("Grayscale Image", grayscale_image)
25         cv2.waitKey(0)
26         cv2.destroyAllWindows()
27
28
29  def main() -> None:
30     input_file = sys.argv[1] if len(sys.argv) > 1 else "input.jpg"
31     show_windows = "--show" in sys.argv[2:] or "--show" in sys.argv[1:2]
32     process_image(str(Path(input_file)), show_windows=show_windows)
33
34
35  if __name__ == "__main__":
36     main()
37
```

## **INPUT**



## **OUTPUT**



## **RESULT**

**The input image was successfully read using Python, and the color image was converted into a grayscale image and displayed successfully.**